

Dot Product

$$X_1 = [x_{11} \ x_{12} \ \dots \ x_{1n}]^T \in \mathbb{R}^n$$

$$X_2 = [x_{21} \ x_{22} \ \dots \ x_{2n}]^T \in \mathbb{R}^n \Rightarrow X_1 \cdot X_2 = x_{11} \cdot x_{21} + x_{12} \cdot x_{22} + \dots + x_{1n} \cdot x_{2n}$$

$$X_1 \cdot X_2 = X_1^T \cdot X_2$$

Angle between vector

$$\langle a, b \rangle = \|a\| \|b\| \cos(\vec{a}, \vec{b})$$

Norm: L1 & L2

$$L_1 = \sum_{i=1}^n |x_i|$$

Manhattan Norm

$$L_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

Euclidean Norm

Distance between 2 Vectors

$$\|x - y\| = \sqrt{\langle x - y, x - y \rangle}$$

Projection of v on u

$$\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u$$

Length of a Vector

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x^T A x}$$

$$\|x\|_2 = \sqrt{x \cdot x} = \sqrt{x^T \cdot x}$$

Ngang trước
dọc sau*

BiLinear
(song
tuyến
tính)

is called bilinear (song tuyến tính) if.

$$i) \Omega(x, \alpha y + \beta z) = \alpha \Omega(x, y) + \beta \Omega(x, z)$$

$$ii) \Omega(\alpha x + \beta y, z) = \alpha \Omega(x, z) + \beta \Omega(y, z)$$

**Inner
product**

Symmetric

is symmetric if $\Omega(x, y) = \Omega(y, x)$

**Positive
Definite**

is positive definite (xác định dương) if

$$\Omega(x, x) \geq 0 \quad \forall x \in \mathbb{R}^n$$

$$\text{and } '= ' \Leftrightarrow x = \vec{0}$$

Orthogonal
(Trực giao)

$$\{x_1, x_2, \dots, x_n\} = \text{orthogonal} \quad \text{if} \quad \boxed{x_i \perp x_j \quad \forall i \neq j}$$
$$\Leftrightarrow \boxed{x_i \cdot x_j = 0}$$

Orthonormal
(Trực chuẩn)

$$\{x_1, x_2, \dots, x_n\} = \text{orthonormal} \quad \text{if} \quad \boxed{\begin{cases} x_i \cdot x_j = 0 \quad \forall i \neq j \\ \|x_i\|_2 = 1 \quad \forall i \end{cases}}$$

Gram-schmidt
Process

$$\text{proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \cdot \vec{v}$$

⊗ $x_1 = x_1$

$$\rightarrow \underline{y_k^i = x_k - \text{proj}_{y_1}(x_k) - \text{proj}_{y_2}(x_k) - \dots - \text{proj}_{y_{k-1}}(x_k)}$$

Orthogonal
matrix (Ma
trận trực giao)

$$\overbrace{A \in \mathbb{R}^{n \times n}} \text{ is orthogonal}$$
$$\text{if} \quad \boxed{A A^T = I \quad \Leftrightarrow \quad A^T = A^{-1}}$$

**Projection
on
Subspace**

⊗

$$U = \text{span}\{b_1, b_2, \dots, b_n\}$$

$$X \in U \Rightarrow X = \lambda_1 b_1 + \lambda_2 b_2 + \dots + \lambda_n b_n$$

$$= B(B^T B)^{-1} B^T X$$

Chap 4

**Determinant
(Định thức)**

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \det A = |A| = a \cdot d - b \cdot c$$

2 x 2

$$|A| = + (a_{11} a_{22} a_{33} + a_{21} a_{32} a_{13} + a_{12} a_{23} a_{31})$$

3 x 3

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} - (a_{31} a_{22} a_{13} + a_{12} a_{21} a_{33} + a_{11} a_{23} a_{32})$$

**Properties Of
Determinant**

$$|A^T| = |A|$$

$$|A+B| = |A| + |B|$$

$$|A \cdot B| = |A| \cdot |B|$$

$$|KA| = K^n |A|$$

$$\det A = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$$

Trace

Tổng ptử đường chéo chính

$$\text{trace}(A) = \sum_{i=1}^n a_{ii} \quad i, j = 1, 2, \dots, n$$

$$\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$$

Properties of Trace

$$\text{tr}(A) = \text{tr}(A^T)$$

$$\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$$

$$\text{tr}(A \cdot B) = \text{tr}(B \cdot A)$$

Eigen Value, Eigen Vector

Ind eigen value

$$|A - \lambda I| = 0$$

$$A \cdot X = \lambda X$$

λ = eigen value

X = eigen vector

Chracteristic Polynomial

$$|A - \lambda I|$$

Đa thức đặc trưng:

$$P_A(\lambda) = \det(A - \lambda I) = -\lambda^3 + a\lambda^2 + b\lambda + c$$

Khi đó:

1. Với $\lambda = 0$ thì $P_A(0) = \det A = c$.
2. Với $\lambda = 1$ thì $P_A(1) = \det(A - I) = -1 + a + b + c$.

Từ đó ta có:

1. $a = \text{trace}(A)$.
2. $c = \det A$.
3. $b = \det(A - I) + 1 - a - c$

CT tính nhanh eigenvalues cho MT 3x3

**Eigen
Decomposition**

$$A = P D P^{-1}$$

$$+) \quad D = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & \lambda_n \end{bmatrix} \quad P = [x_1 \dots x_n]$$

$\downarrow$
eigen vector of A

$$\begin{aligned} \rightarrow A^{100} &= P D P^{-1} P D P^{-1} P D P^{-1} \dots P D P^{-1} \\ &= P D^{100} P^{-1} \end{aligned}$$

Single Value Decomposition (SVD)

$$A = U \Sigma V^T$$

Diagram illustrating the dimensions of the matrices in SVD:

- A is $m \times n$.
- U is $m \times m$.
- Σ is $m \times n$.
- V^T is $n \times n$.

$$U = [u_1, \dots, u_n] = \text{orthogonal matrix}$$

$$\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n & \dots & 0 \end{bmatrix}$$

$$V = [v_1, \dots, v_n] = \text{orthogonal matrix}$$

* summary:

$$\underline{A = U \Sigma V^T}$$

$$B_1: \underline{A^T A} = ?$$

B_2 : Find eigen value, eigen vector of $\underline{A^T A}$

$$\lambda = \lambda_1, \dots, \lambda_n$$

$$x = x_1, \dots, x_n$$

$$B_3: \sigma_n = \sqrt{\lambda_n} \Rightarrow \Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix}$$

$$V_n = \frac{x_n}{\|x_n\|_2} \Rightarrow V = [v_1, \dots, v_n] \Rightarrow B_4$$

$$B_5: u_n = \frac{A v_n}{\sigma_n} \Rightarrow U = [u_1, \dots, u_n]$$

Rank K
approximate
of A

rank k of A

$$\hat{A}(k) = \sigma_1 \bar{u}_1 \bar{v}_1^T + \dots + \sigma_k \bar{u}_k \bar{v}_k^T$$

$$\hat{A}(1) = \sigma_1 \bar{u}_1 \bar{v}_1^T \quad \hat{A}(2) = \sigma_1 \bar{u}_1 \bar{v}_1^T + \sigma_2 \bar{u}_2 \bar{v}_2^T$$

$$\|A - \hat{A}(k)\|_2 = \sigma_{k+1}$$

CT Nhanh

Chap 5

Derivative

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$(x^a)' = \frac{1}{x^{1-a}}$$

$$(x^2)' = 2x \quad (x^n)' = nx^{n-1}$$

$$(e^x)' = e^x \quad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

*Some derivative formula

Linear
Approximate

$$y = f(a) + f'(a)(x-a)$$

Taylor Approximate

$$f(x) = a_0 + a_1(x-a) + a_2(x-a)^2 + \dots + a_n(x-a)^n$$

with $a_k = \frac{f^{(k)}(a)}{k!}$

$k! = 1 \cdot 2 \cdot \dots \cdot k$
 $1! = 1$
 $2! = 1 \cdot 2$
 $3! = 1 \cdot 2 \cdot 3$

$a_0 = f(a)$

Maclaurin Series

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

When $a=0$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$$

*Taylor series nma
 $a = 0$

Chain rule

multi-variate : $f = f(x_1, x_2)$
 $x_1 = x_1(t)$
 $x_2 = x_2(t)$

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \frac{dx_1}{dt} + \frac{\partial f}{\partial x_2} \frac{dx_2}{dt} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix}$$

uni-variate : $f = f(x(t))$
 $f' = \frac{df}{dx} \cdot \frac{dx}{dt}$

Jacobian Matrix

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \text{Jacobian matrix}$$

$\uparrow$
 $n \times 2$

Hessian Matrix

$$\frac{\partial^2 f}{\partial x^2} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \dots & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \Rightarrow \text{Hessian matrix}$$

* Degree 2: $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $x_0 = [a \ b]$

$$f(x, y) = f(x_0) + \frac{\partial f}{\partial x}(x-a) + \frac{\partial f}{\partial y}(y-b) + \frac{1}{2!} \left[\frac{\partial^2 f}{\partial x^2}(x-a)^2 + 2 \frac{\partial^2 f}{\partial x \partial y}(x-a)(y-b) + \frac{\partial^2 f}{\partial y^2}(y-b)^2 \right]$$

Taylor for multi-variate function

PMF
(Probability
Mass
Function)

$$P(X=k) = p(k)$$

$$(1) \quad 0 \leq p(k) \leq 1$$

$$(2) \quad \sum p(k) = 1$$

X : discrete

PDF
(Probability
Density
Function)

$$P(X \leq k) = \int_{-\infty}^k p(x) dx$$

$$(1) \quad p(x) \geq 0$$

$$(2) \quad \int_{-\infty}^{+\infty} p(x) dx = 1$$

X : continuous

Multiple
Integral
(Tích phân
bội)

$$\int_a^b \int_c^d f(x, y) dx dy = \int_a^b \left[\int_c^d f(x, y) dy \right] dx$$

X Discrete

Joint
Distribution

$$P(X = x_i, Y = y_j) = f(x_i, y_j)$$

$$(1) 0 \leq f(x_i, y_j) \leq 1$$

$$(2) \sum f(x_i, y_j) = 1$$

Margin
Probability

$$f(x) = \text{PMF of } X$$

$$P(X = x_i) = \sum_{j=1}^m P(X = x_i, Y = y_j) \\ = \sum_{j=1}^m f(x_i, y_j)$$

Conditional
Probability

$$X, Y = \text{discrete} : X = \{x_1, \dots, x_n\} \quad Y = \{y_1, \dots, y_m\} \\ P(X = x_i | Y = y_j) = \frac{P(X = x_i, Y = y_j)}{P(Y = y_j)}$$

X Continuous

$$P(X \leq a, Y \leq b) = \int_{-\infty}^a \int_{-\infty}^b f(x, y) dy dx$$

$$(1) f(x, y) \geq 0$$

$$(2) \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$$

$$P(X) = \int_{-\infty}^{+\infty} f(x, y) dy$$

$$P(Y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

X Discrete

X Continuous

Mean

$$E(xy)$$

$$\sum xy P(x,y)$$

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy P(x,y) dx dy$$

$$E(x)$$

$$\sum x P(x)$$

$$\int x P(x) dx$$

Variance

$$V(x)$$

$$E(x^2) - (E(x))^2$$

$$\int_{-\infty}^{+\infty} (x - \mu)^2 P(x) dx$$

$$E(x^2) - (E(x))^2$$

Covariance

$$\text{cov}(x,y)$$

$$E(xy) - E(x) \cdot E(y)$$

$$\text{cov}(x,x) = V(x)$$

$$= E(x^2) - (E(x))^2$$

X Discrete

X Continuous

Correlation

$$\text{corr}(x, y) = \frac{\text{cov}(x, y)}{\sqrt{V(x) \cdot V(y)}}$$

Correlation matrix

$$\begin{aligned} \text{cov}(w, w) &= E(w w^T) - E(w)E(w^T) \\ &= \begin{bmatrix} \text{cov}(x_1, x_1) & \text{cov}(x_1, x_2) & \dots & \text{cov}(x_1, x_D) \\ \text{cov}(x_2, x_1) & \dots & \dots & \dots \\ \vdots & & & \\ \text{cov}(x_D, x_1) & \dots & \dots & \text{cov}(x_D, x_D) \end{bmatrix} \in \mathbb{R}^{D \times D} \end{aligned}$$

Dataset

$$w_1, w_2, \dots, w_D \in \mathbb{R}^D$$

Mean

$$\bar{w} = \frac{1}{n} (w_1 + w_2 + \dots + w_D) \in \mathbb{R}^D$$

Dataset

$$w_1, w_2, \dots, w_D \in \mathbb{R}^D$$

Covariance

$$\text{cov} = \frac{1}{n} \sum (w_i - \bar{w})(w_i - \bar{w})^T \in \mathbb{R}^D$$

Properties of
Covariance

$$- E(w_1 + w_2) = E(w_1) + E(w_2)$$

$$- E(\alpha w_1) = \alpha E(w_1)$$

$$- V(\alpha w_1) = \alpha^2 V(w_1)$$

$$- \underline{V(w_1 + w_2)} = V(w_1) + V(w_2) + \text{cov}(w_1, w_2) + \text{cov}(w_2, w_1)$$

$$\text{if } x, y \text{ is random variable} \Rightarrow \underline{\text{cov}(w_1, w_2) = \text{cov}(w_2, w_1)}$$

$x, y =$
Independent

$$\Leftrightarrow P(x|y) = P(x)$$

$$\Leftrightarrow P(x, y) = P(x) P(y)$$

Normal Distribution

$$X \sim N(\mu, \sigma^2) \sim N(X | \mu, \sigma^2)$$

X = random variable

$$P(X) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

Standard Normal Distribution

$$X \sim N(0, \underline{1}) \quad (\text{chuẩn tắc})$$

Multivariate Normal Distribution

$w = [x \ y]^T$ has normal distribution with mean $\mu_w = [\mu_x \ \mu_y]^T$ and covariance matrix $\Sigma = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix}$

is multivariate if

$$P(x, y) = \frac{1}{2\pi \sqrt{\det(\Sigma)}} e^{-\frac{1}{2} (w - \mu_w)^T \Sigma^{-1} (w - \mu_w)}$$

**Marginal
Normal of
Distribution**

$$\begin{bmatrix} X & Y \end{bmatrix}^T = \underline{N(\mu, \Sigma)}$$

$$\underline{\mu} = [\mu_x \quad \mu_y]^T$$
$$\underline{\Sigma} = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix}$$

$$\Rightarrow X = N(\mu_x, \Sigma_{xx})$$

$$Y = N(\mu_y, \Sigma_{yy})$$

**Conditional
Normal of
Distribution**

$$P(X|Y=y) = N(\mu_{X|Y=y}, \Sigma_{X|Y=y})$$

$$\mu_{X|Y=y} = \mu_x + \Sigma_{xy} \Sigma_{yy}^{-1} (y - \mu_y)$$

$$\Sigma_{X|Y=y} = \Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}$$

**Sum of 2
Normal of
Distribution**

$$W_1 = N(\mu_1, \Sigma_1) \quad , \quad W_2 = N(\mu_2, \Sigma_2)$$

$$aW_1 + bW_2 = N(a\mu_1 + b\mu_2, \quad a^2\Sigma_1 + b^2\Sigma_2)$$

Gradient Descent

$$\begin{aligned} [x_n \ y_n]^T &= [x_{n-1} \ y_{n-1}]^T - \eta_{n-1} [\nabla f(x_{n-1}, y_{n-1})]^T \\ &= [x_{n-1} \ y_{n-1}]^T - \eta_{n-1} \left[\frac{\partial f}{\partial x}(x_{n-1}, y_{n-1}), \frac{\partial f}{\partial y}(x_{n-1}, y_{n-1}) \right] \end{aligned}$$

Gradient Descent with Momentum

$$\begin{aligned} [x_n \ y_n]^T &= [x_{n-1} \ y_{n-1}]^T - \eta_{n-1} [\nabla f(x_{n-1}, y_{n-1})]^T \\ &\quad + \alpha \Delta x \end{aligned}$$

Constrain Optimization

$$\begin{aligned} &\text{Min } f(x) \quad x \in \mathbb{R}^n \\ &\text{subject to: } g_1(x) \leq 0 \\ &\quad \vdots \\ &\quad g_m(x) \leq 0 \end{aligned}$$

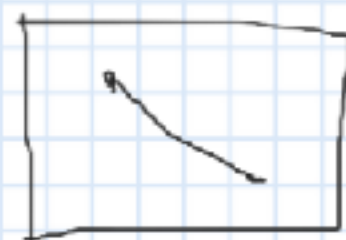
(bi-ràng)
biên giới

Lagrangian duality function

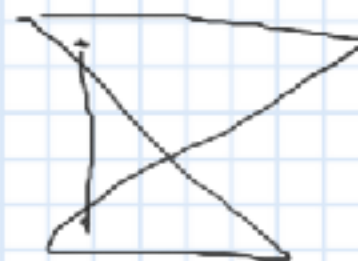
$$\begin{aligned} \mathcal{L}(x, \lambda) &= f(x) + \lambda_1 g_1(x) + \dots + \lambda_m g_m(x) \\ &= f(x) + \lambda^T g(x) \end{aligned}$$

Convex set

* tập giá trị [0,1]:



convex



not convex

* convex set : $D \subseteq \mathbb{R}^n$ is convex
(tập lồi)
 $\Rightarrow \forall A, B \in D : \theta A + (1 - \theta)B \in D$
 $\forall 0 \leq \theta \leq 1$

Convex function

$$f(\theta A + (1 - \theta)B) \leq \theta f(A) + (1 - \theta)f(B)$$

Hessian x
Convex
function

* theorem: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex function
if Hessian matrix is semi-definite matrix
($x^T A x \geq 0 \forall x$)

Linear Programming

$$\begin{aligned} \min: & c_1 x_1 + \dots + c_n x_n = c^T x \\ \text{s.t.} : & \begin{cases} a_{11} x_1 + \dots + a_{1n} x_n \leq b_1 \\ \vdots \\ a_{m1} x_1 + \dots + a_{mn} x_n \leq b_m \end{cases} \quad \Leftrightarrow Ax \leq b \end{aligned}$$

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \end{aligned} \Rightarrow \text{Lagrangian duality}$$

Quadratic Programming

$$\begin{aligned} \min \quad & \frac{1}{2} x^T Q x + c^T x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

$$\Rightarrow \text{duality: } \max -\frac{1}{2} [c^T + \lambda^T A] Q^{-1} [c + A^T \lambda] - \lambda^T b$$